

# Classification of Universal Forms over Number Fields

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# Quadratic forms

A **quadratic form over  $\mathbb{Z}$**  is

$$Q(x_1, x_2, \dots, x_n) = \sum_{1 \leq i \leq j \leq n} a_{ij} x_i x_j, \quad a_{ij} \in \mathbb{Z}.$$

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*Which integers are represented by  $Q$ ?*

## Examples

- Pythagorean triple:  $x^2 + y^2 - z^2 = 0$
- Fermat:  $x^2 + y^2 = p \iff p \equiv 1 \pmod{4}$
- Legendre:  $x^2 + y^2 + z^2 = n \iff n \neq 4^k(8m+7)$
- **Lagrange:  $x^2 + y^2 + z^2 + w^2 = n$  for all  $n \in \mathbb{Z}_{\geq 0}$ .**

# Universal quadratic forms

- **Positive definite:**  $Q(x) > 0$  for all  $x \neq 0$ .
- **Universal quadratic forms:** Positive definite quadratic forms representing all positive integers.

## Classifications

- **Ramanujan (1917), Dickson (1927):** exactly 54 universal diagonal quaternary forms  $ax^2 + by^2 + cz^2 + dw^2$ .
- **15-Theorem (Conway–Schneeberger, 1993):** **Classical** quadratic form is universal if and only if it represents 1, 2, 3, 5, 6, 7, 10, 14, and 15.

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## 290-Theorem (Bhargava–Hanke, 2011)

If a positive definite quadratic form  $Q$  (over  $\mathbb{Z}$ ) represents

1, 2, 3, 5, 6, 7, 10, 13, 14, 15, 17, 19, 21, 22, 23, 26,  
29, 30, 31, 34, 35, 37, 42, 58, 93, 110, 145, 203, and 290,

then it is universal.

# Criterion sets

## Definition (Criterion set)

A finite set  $\mathcal{C}$  is called a **criterion set** if

$$Q \text{ represents all of } \mathcal{C} \iff Q \text{ is universal.}$$

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## $S$ -universal form

Let  $S \subseteq \mathbb{Z}_{\geq 0}$ . We say that a quadratic form  $Q$  is  **$S$ -universal** if it represents all of  $S$ .

## Conjecture (Rouse, 2014)

A positive definite quadratic form represents all positive **odd integers** if and only if it represents  $1, 3, 5, \dots, 451$ .

# Moving to number fields

Let  $K$  be a number field of degree  $d$ , ring of integers  $\mathcal{O}_K$ . A **quadratic form over  $K$**  is

$$Q(x_1, x_2, \dots, x_n) = \sum_{1 \leq i \leq j \leq n} a_{ij} x_i x_j, \quad a_{ij} \in \mathcal{O}_K.$$

- $K$  **totally real**: all embeddings  $\sigma_1, \dots, \sigma_d : K \rightarrow \mathbb{R}$ .
- $\alpha \succ 0$  **totally positive**:  $\sigma_i(\alpha) > 0$  for all  $i$ .  $\mathcal{O}_K^+$  is the set of all totally positive integers in  $\mathcal{O}_K$ .
- $Q$  **totally positive definite**: every  $\sigma_i(Q)$  positive definite.
- $Q$  **represents**  $\alpha \in \mathcal{O}_K^+$  if there exists  $x \in \mathcal{O}_K^n$  with  $Q(x) = \alpha$ .
- $Q$  **universal**: represents every  $\alpha \in \mathcal{O}_K^+$ .
- $\alpha \in \mathcal{O}_K^+$  is represented by  $Q \iff \alpha \varepsilon^2$  is represented by  $Q$  for all  $\varepsilon \in \mathcal{O}_K^\times$ . Enough to consider representation of  $\mathcal{O}_K^+ / \mathcal{O}_K^{\times 2}$ .

# Do universal forms exist over $K$ ?

- **Maaß (1941)**: over  $\mathbb{Q}(\sqrt{5})$ , already  $x^2 + y^2 + z^2$  is universal!
- **Siegel (1945)**: sums of squares (any number!) are universal *only* over  $\mathbb{Q}$  and  $\mathbb{Q}(\sqrt{5})$ .
- **Hsia–Kitaoka–Kneser (1978)**: universal quadratic forms exist over *every* totally real field.

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## Kitaoka conjecture

Only a finitely many  $K$  admits a **ternary** universal quadratic forms.

Known: **odd degree**  $K$  (Earnest-Khosravani, 1997), **any fixed degree** (Kala–Yatsyna, 2021)

- **Blomer–Kala (2015)**: minimal rank can be arbitrarily large.

# Criterion sets over number fields

- **Chan–Oh (2023)**: finite criterion sets exist over every totally real  $K$ .
- **Lee (2008)**: the only explicitly known beyond  $\mathbb{Q}$  – over  $\mathbb{Q}(\sqrt{5})$ , a classical form is universal iff it represents

$$1, \quad 2, \quad \frac{5+\sqrt{5}}{2}, \quad \frac{7\pm\sqrt{5}}{2}, \quad 2 \cdot \frac{5+\sqrt{5}}{2}, \quad 3 \cdot \frac{5+\sqrt{5}}{2}.$$

- **Kala–Krásenský–Romeo (2026)**: there exists a **unique** criterion set  $\mathcal{C}_K$  which **minimal** w.r.t inclusion. Moreover,  $\mathcal{C}_K$  consists precisely of *critical* integers.

*We don't know any explicit  $\mathcal{C}_K$ !*

# Higher degree forms ( $m$ -ic forms)

An  $m$ -ic form in  $n$  variables over  $K$  is a homogeneous polynomial of degree  $m > 2$ .

$$Q(x_1, \dots, x_n) = \sum_{\substack{i_1, \dots, i_n \geq 0 \\ i_1 + \dots + i_n = m}} a_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n}, \quad a_{i_1 \dots i_n} \in \mathcal{O}_K.$$

**Motivation (Waring's problem):** every positive integer is a sum of at most  $g(m) < \infty$   $m$ th powers (Hilbert, 1909). Hence

$$x_1^m + x_2^m + \cdots + x_{g(m)}^m$$

is a universal (diagonal)  $m$ -ic form over  $\mathbb{Z}$ .

- Examples of universal  $m$ -ic forms:  $g(4) = 19, g(6) = 73$  etc.
- Doesn't work for  $K$ .

# Universal $m$ -ic forms

- **Obstruction:** (say) over  $\mathbb{Q}(\sqrt{2})$ ,

$$(a + b\sqrt{2})^2 = a^2 + 2b^2 + 2ab\sqrt{2}.$$

- **Siegel:** for every  $m > 2$ , sums of  $m$ th powers are never universal over  $K \neq \mathbb{Q}$ .
- Waring (circle method) over  $K$  (Siegel, Birch, Ramanujam) holds *inside the subring generated by  $m$ -th powers*, for elements of large norm.

So it is genuinely unclear whether universal  $m$ -ic forms over  $K$  exist at all...

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## Theorem (Kala–P., 2024)

For every totally real  $K$  and every even  $m > 2$ , a **universal  $m$ -ic form over  $K$  exists**.

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What about  $\mathcal{C}_K$ ?

# No finite criterion set

Perhaps surprisingly, the answer is **NO** – even over  $K = \mathbb{Q}$ .

## Proposition (Kala-P., 2024)

Given a positive even integer  $m > 2$  and a positive integer  $B$ , there is a positive definite  $m$ -ic form  $Q$  that represents all the positive integers  $\leq B$  but is **not** universal.

### Proof.

- $Q_1(x_1, \dots, x_B) = \sum_{i=1}^B i x_i^m$ .
- Let  $c > B$  be  $m$ th powerfree.
- $Q(x_1, \dots, x_B) = \sum_{i=1}^B i x_i^m + \sum_{1 \leq i < j \leq B} (c+1) x_i^2 x_j^{m-2}$ .

$m > 2$  is important.

# All critical elements

Classified all infinitely many critical elements.

## Theorem (Kala-P., 2024)

Let  $\mathcal{A} \subset \mathbb{Z}_{>0}$  be finite and  $m \in \mathbb{Z}_{>2}$  even. Then the following conditions are equivalent:

- 1 There exists a positive definite  $m$ -ic form  $Q$  that represents exactly  $\mathbb{Z}_{\geq 0} \setminus \mathcal{A}$ .
- 2 For all  $a, b \in \mathbb{Z}$ , we have that  $ab^m \in \mathcal{A}$  implies  $a \in \mathcal{A}$ .

In particular, every  $m$ th powerfree integer is a critical element.

## Theorem (Kala-P., 2024)

Similar classification of all critical elements over  $K$ .

*What is so special about  $m = 2$ ?*

# How the 290-Theorem is proved: escalation

Where do the magic numbers 15 and 290 come from?

## Truant

The **truant** of a form  $Q$  is the *smallest* positive integer that  $Q$  fails to represent.

## Escalation

If  $Q$  has truant  $t$ , an **escalation** of  $Q$  is any positive definite form obtained by adding *one new variable* so that the new form represents  $t$ .

Start from the “empty” form of rank 0 (it represents only 0, so its truant is 1) and escalate repeatedly: this grows a **tree** of forms, one level per variable. Universal forms are the leaves where the process stops.

# Escalation over $\mathbb{Z}$ : the tree is finite

**Step 0:** the zero form has truant 1  $\Rightarrow$  every escalation contains  $x^2$ .

**Step 1:**  $x^2$  has truant 2  $\Rightarrow$  escalate to

$$x^2 + cxy + 2y^2, \quad \text{pos. def.} \iff c^2 < 4 \cdot 1 \cdot 2 \Rightarrow c \in \{0, \pm 1, \pm 2\}.$$

## The key finiteness

Cross coefficients are bounded by Cauchy–Schwarz:  $|c| \leq 2\sqrt{at}$ .  
 $\Rightarrow$  every node of the tree has only **finitely many** forms.

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 $\Rightarrow$  every node of the tree has only **finitely many** forms.

## For $m$ -ic forms—This is not true!

Given positive integers  $a, b$ , there are **infinitely** many positive definite forms representing both  $a, b$ :  $ax^m + cx^{m-2}y^2 + by^m$  for all  $c \in \mathbb{Z}_{\geq 0}$ .

# Diagonal forms

What if we forbid cross terms?

Diagonal  $m$ -ic forms

$$Q(x_1, \dots, x_n) = a_1 x_1^m + \dots + a_n x_n^m, \quad a_i \in \mathcal{O}_K^+.$$

*What about  $\mathcal{C}_K$  for these forms? – Exists!*

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What about  $\mathcal{C}_K$  for these forms? – **Exists!**

## Escalation over $K$ :

- **Truant:** We say that  $\alpha \in \mathcal{O}_K^+ / \mathcal{O}_K^{\times m}$  is a truant of  $Q$  if  $Q$  does not represent  $\alpha$ , and  $Q$  represents all  $\beta \in \mathcal{O}_K^+ / \mathcal{O}_K^{\times m}$  with  $N(\beta) < N(\alpha)$ . Denote by  $T(Q)$  the set of all truants.
- **Escalation:** Assume that  $Q$  is non-universal with a truant  $\alpha$ . An escalation of  $Q$  at  $\alpha$  is a totally positive definite diagonal  $m$ -ic form  $Q' = Q \perp \langle \beta \rangle$  with  $\beta \in \mathcal{O}_K^+ / \mathcal{O}_K^{\times m}$  such that  $Q'$  represents  $\alpha$ .

$$E(Q, \alpha) = \{\beta \in \mathcal{O}_K^+ / \mathcal{O}_K^{\times m} \mid Q \perp \langle \beta \rangle \text{ represents } \alpha\}$$

$$E(Q) = \bigcup_{\alpha \in T(Q)} E(Q, \alpha)$$

## Proposition (P., 2026+)

- 1  $T(Q)$  is finite.
- 2 For each  $\alpha \in T(Q)$ , the set  $E(Q, \alpha) \neq \emptyset$  is finite.
- 3  $E(Q)$  is finite.
- 4 If this process **terminates**, then a finite  $\mathcal{C}_K$  exists for diagonal  $m$ -ic forms.

# Local–global principle

- Local–global can **fail** for higher degree.
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## Theorem (P., 2026+)

Let  $m \geq 2$  be an even integer,  $K$  be a totally real number field of degree  $d$ , and  $Q(x) = \sum_{i=1}^n a_i x_i^m$  be a totally positive definite diagonal  $m$ -ic form in  $n$  variables over  $K$  with  $n > d(d+1)(m-1)2^{m-1}$ .

Assume that there exists an integer  $t \geq 0$  such that for every  $\alpha \in \mathcal{O}_K^+$  and for every prime ideal  $\mathfrak{p} \subset \mathcal{O}_K$  there exists  $x \in \mathcal{O}_{\mathfrak{p}}^n$  with  $Q(x) = \alpha$  and

$$\min_{1 \leq i \leq n} v_{\mathfrak{p}}(x_i) \leq t.$$

Then, there exists a constant  $C > 0$  such that every  $\alpha \in \mathcal{O}_K^+$  with  $N(\alpha) > C$  is represented by  $Q$ .

# Proof strategy—very rough

- We need to count  $V_K(\alpha) = \# \{x \in \mathcal{O}_K^n \mid Q(x) = \sum_{i=1}^n a_i x_i^m = \alpha\}$ .
- **Weil restriction**:  $V_{\mathbb{Z}}(\alpha) = \# \{x \in \mathbb{Z}^{nd} \mid f_i(x) = \alpha_i \forall 1 \leq i \leq d\}$ .
- We count:

$$M(P; \alpha) = \#\{x \in \mathbb{Z}^{nd} \cap P\mathcal{B} \mid f_i(x) = \alpha_i \forall 1 \leq i \leq d\}, \quad \mathcal{B} \subset [-1, 1]^{nd}$$

- Using **Birch**:

$$M(P; \alpha) = P^{d(n-m)} \cdot \mathfrak{S} \cdot \Psi + O(P^{d(n-m)-\delta})$$

- The main term can easily be zero! – need careful treatment of  $\mathfrak{S}, \Psi$ .

Consequently, the escalation algorithm **terminates**  $\implies$  finite criterion set  $\mathcal{C}_K$ !

# Finally

Characterization of critical elements:

## Proposition (P., 2026+)

Let  $m \geq 2$  be an even integer and  $K$  be a totally real number field. Then,  $\alpha \in \mathcal{O}_K^+ / \mathcal{O}_K^{\times m}$  is critical if and only if  $\alpha$  is a truant of some totally positive definite diagonal  $m$ -ic form over  $K$ .

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**Thank You for Your Attention!**

Questions?